# MSG No. 210.53 $F4_132'$ [ Type II, cubic ]

Table 1: Wyckoff site: 8a, site symmetry: 23.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | [1,8,9,10,17,18,19,20,21,22,23,24,97,104,105,106,113,114,115,116,117,118,119,120] |
| 2 | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [2,3,4,5,6,7,11,12,13,14,15,16,98,99,100,101,102,103,107,108,109,110,111,112] |
| 3 | $[0, \frac{1}{2}, \frac{1}{2}]$ | [25,32,33,34,41,42,43,44,45,46,47,48,121,128,129,130,137,138,139,140,141,142,143,144] |
| 4 | $[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$ | [26,27,28,29,30,31,35,36,37,38,39,40,122,123,124,125,126,127,131,132,133,134,135,136] |
| 5 | $[\frac{1}{2}, 0, \frac{1}{2}]$ | [49,56,57,58,65,66,67,68,69,70,71,72,145,152,153,154,161,162,163,164,165,166,167,168] |
| 6 | $[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$ | [50,51,52,53,54,55,59,60,61,62,63,64,146,147,148,149,150,151,155,156,157,158,159,160] |
| 7 | $[\frac{1}{2}, \frac{1}{2}, 0]$ | [73,80,81,82,89,90,91,92,93,94,95,96,169,176,177,178,185,186,187,188,189,190,191,192] |
| 8 | $[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$ | [74,75,76,77,78,79,83,84,85,86,87,88,170,171,172,173,174,175,179,180,181,182,183,184] |

Table 2: Wyckoff site: 8b, site symmetry: 23.1'

| No. | position | mapping |
|---|---|---|
| 1 | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [1,8,9,10,17,18,19,20,21,22,23,24,97,104,105,106,113,114,115,116,117,118,119,120] |
| 2 | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [2,3,4,5,6,7,11,12,13,14,15,16,98,99,100,101,102,103,107,108,109,110,111,112] |
| 3 | $[\frac{1}{2}, 0, 0]$ | [25,32,33,34,41,42,43,44,45,46,47,48,121,128,129,130,137,138,139,140,141,142,143,144] |
| 4 | $[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$ | [26,27,28,29,30,31,35,36,37,38,39,40,122,123,124,125,126,127,131,132,133,134,135,136] |
| 5 | $[0, \frac{1}{2}, 0]$ | [49,56,57,58,65,66,67,68,69,70,71,72,145,152,153,154,161,162,163,164,165,166,167,168] |
| 6 | $[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$ | [50,51,52,53,54,55,59,60,61,62,63,64,146,147,148,149,150,151,155,156,157,158,159,160] |
| 7 | $[0, 0, \frac{1}{2}]$ | [73,80,81,82,89,90,91,92,93,94,95,96,169,176,177,178,185,186,187,188,189,190,191,192] |
| 8 | $[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$ | [74,75,76,77,78,79,83,84,85,86,87,88,170,171,172,173,174,175,179,180,181,182,183,184] |

Table 3: Wyckoff site: 16c, site symmetry: .321'

| No. | position | mapping |
|---|---|---|
| 1 | $[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$ | [1,12,14,16,17,18,97,108,110,112,113,114] |
| 2 | $[\frac{3}{8}, \frac{1}{8}, \frac{3}{8}]$ | [2,7,15,57,67,72,98,103,111,153,163,168] |
| 3 | $[\frac{3}{8}, \frac{3}{8}, \frac{1}{8}]$ | [3,4,11,82,92,93,99,100,107,178,188,189] |
| 4 | $[\frac{1}{8}, \frac{3}{8}, \frac{3}{8}]$ | [5,6,13,32,46,47,101,102,109,128,142,143] |
| 5 | $[\frac{1}{8}, \frac{7}{8}, \frac{7}{8}]$ | [8,22,23,29,30,37,104,118,119,125,126,133] |
| 6 | $[\frac{7}{8}, \frac{1}{8}, \frac{7}{8}]$ | [9,19,24,50,55,63,105,115,120,146,151,159] |
| 7 | $[\frac{7}{8}, \frac{7}{8}, \frac{1}{8}]$ | [10,20,21,75,76,83,106,116,117,171,172,179] |
| 8 | $[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}]$ | [25,36,38,40,41,42,121,132,134,136,137,138] |
| 9 | $[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$ | [26,31,39,81,91,96,122,127,135,177,187,192] |
| 10 | $[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$ | [27,28,35,58,68,69,123,124,131,154,164,165] |
| 11 | $[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$ | [33,43,48,74,79,87,129,139,144,170,175,183] |
| 12 | $[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$ | [34,44,45,51,52,59,130,140,141,147,148,155] |
| 13 | $[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}]$ | [49,60,62,64,65,66,145,156,158,160,161,162] |
| 14 | $[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$ | [53,54,61,80,94,95,149,150,157,176,190,191] |
| 15 | $[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$ | [56,70,71,77,78,85,152,166,167,173,174,181] |
| 16 | $[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}]$ | [73,84,86,88,89,90,169,180,182,184,185,186] |

Table 4: Wyckoff site: `16d`, site symmetry: `.321'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$ | `[1,12,14,16,17,18,97,108,110,112,113,114]` |
| 2 | $\left[\frac{7}{8}, \frac{5}{8}, \frac{7}{8}\right]$ | `[2,7,15,57,67,72,98,103,111,153,163,168]` |
| 3 | $\left[\frac{7}{8}, \frac{7}{8}, \frac{5}{8}\right]$ | `[3,4,11,82,92,93,99,100,107,178,188,189]` |
| 4 | $\left[\frac{5}{8}, \frac{7}{8}, \frac{7}{8}\right]$ | `[5,6,13,32,46,47,101,102,109,128,142,143]` |
| 5 | $\left[\frac{5}{8}, \frac{3}{8}, \frac{3}{8}\right]$ | `[8,22,23,29,30,37,104,118,119,125,126,133]` |
| 6 | $\left[\frac{3}{8}, \frac{5}{8}, \frac{3}{8}\right]$ | `[9,19,24,50,55,63,105,115,120,146,151,159]` |
| 7 | $\left[\frac{3}{8}, \frac{3}{8}, \frac{5}{8}\right]$ | `[10,20,21,75,76,83,106,116,117,171,172,179]` |
| 8 | $\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$ | `[25,36,38,40,41,42,121,132,134,136,137,138]` |
| 9 | $\left[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}\right]$ | `[26,31,39,81,91,96,122,127,135,177,187,192]` |
| 10 | $\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{8}\right]$ | `[27,28,35,58,68,69,123,124,131,154,164,165]` |
| 11 | $\left[\frac{3}{8}, \frac{1}{8}, \frac{7}{8}\right]$ | `[33,43,48,74,79,87,129,139,144,170,175,183]` |
| 12 | $\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}\right]$ | `[34,44,45,51,52,59,130,140,141,147,148,155]` |
| 13 | $\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$ | `[49,60,62,64,65,66,145,156,158,160,161,162]` |
| 14 | $\left[\frac{1}{8}, \frac{7}{8}, \frac{3}{8}\right]$ | `[53,54,61,80,94,95,149,150,157,176,190,191]` |
| 15 | $\left[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}\right]$ | `[56,70,71,77,78,85,152,166,167,173,174,181]` |
| 16 | $\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$ | `[73,84,86,88,89,90,169,180,182,184,185,186]` |

Table 5: Wyckoff site: `32e`, site symmetry: `.3.1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | `[1,17,18,97,113,114]` |
| 2 | $\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{1}{4}\right]$ | `[2,7,15,98,103,111]` |
| 3 | $\left[x+\frac{1}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$ | `[3,4,11,99,100,107]` |
| 4 | $\left[\frac{1}{4}-x, x+\frac{1}{4}, x+\frac{1}{4}\right]$ | `[5,6,13,101,102,109]` |
| 5 | $[x, -x, -x]$ | `[8,22,23,104,118,119]` |
| 6 | $[-x, x, -x]$ | `[9,19,24,105,115,120]` |
| 7 | $[-x, -x, x]$ | `[10,20,21,106,116,117]` |
| 8 | $\left[\frac{1}{4}-x, \frac{1}{4}-x, \frac{1}{4}-x\right]$ | `[12,14,16,108,110,112]` |
| 9 | $\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[25,41,42,121,137,138]` |
| 10 | $\left[x+\frac{1}{4}, \frac{3}{4}-x, x+\frac{3}{4}\right]$ | `[26,31,39,122,127,135]` |
| 11 | $\left[x+\frac{1}{4}, x+\frac{3}{4}, \frac{3}{4}-x\right]$ | `[27,28,35,123,124,131]` |
| 12 | $\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{3}{4}\right]$ | `[29,30,37,125,126,133]` |
| 13 | $\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[32,46,47,128,142,143]` |
| 14 | $\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[33,43,48,129,139,144]` |
| 15 | $\left[-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[34,44,45,130,140,141]` |
| 16 | $\left[\frac{1}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$ | `[36,38,40,132,134,136]` |
| 17 | $\left[x+\frac{1}{2}, x, x+\frac{1}{2}\right]$ | `[49,65,66,145,161,162]` |
| 18 | $\left[x+\frac{3}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$ | `[50,55,63,146,151,159]` |
| 19 | $\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}-x\right]$ | `[51,52,59,147,148,155]` |
| 20 | $\left[\frac{3}{4}-x, x+\frac{1}{4}, x+\frac{3}{4}\right]$ | `[53,54,61,149,150,157]` |
| 21 | $\left[x+\frac{1}{2}, -x, \frac{1}{2}-x\right]$ | `[56,70,71,152,166,167]` |
| 22 | $\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$ | `[57,67,72,153,163,168]` |
| 23 | $\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$ | `[58,68,69,154,164,165]` |
| 24 | $\left[\frac{3}{4}-x, \frac{1}{4}-x, \frac{3}{4}-x\right]$ | `[60,62,64,156,158,160]` |
| 25 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x\right]$ | `[73,89,90,169,185,186]` |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 26 | $\left[x+\frac{3}{4},\ \frac{3}{4}-x,\ x+\frac{1}{4}\right]$ | [74,79,87,170,175,183] |
| 27 | $\left[x+\frac{3}{4},\ x+\frac{3}{4},\ \frac{1}{4}-x\right]$ | [75,76,83,171,172,179] |
| 28 | $\left[\frac{3}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$ | [77,78,85,173,174,181] |
| 29 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$ | [80,94,95,176,190,191] |
| 30 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$ | [81,91,96,177,187,192] |
| 31 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$ | [82,92,93,178,188,189] |
| 32 | $\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$ | [84,86,88,180,182,184] |

Table 6: Wyckoff site: 48f, site symmetry: 2..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | [1,8,97,104] |
| 2 | $\left[x+\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$ | [2,3,98,99] |
| 3 | $\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}-x\right]$ | [4,16,100,112] |
| 4 | $\left[\frac{1}{4},\ \frac{1}{4},\ x+\frac{1}{4}\right]$ | [5,15,101,111] |
| 5 | $\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{4}\right]$ | [6,11,102,107] |
| 6 | $\left[\frac{1}{4},\ \frac{1}{4}-x,\ \frac{1}{4}\right]$ | [7,12,103,108] |
| 7 | $[-x,\ 0,\ 0]$ | [9,10,105,106] |
| 8 | $\left[\frac{1}{4}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$ | [13,14,109,110] |
| 9 | $[0,\ x,\ 0]$ | [17,24,113,120] |
| 10 | $[0,\ 0,\ x]$ | [18,21,114,117] |
| 11 | $[0,\ 0,\ -x]$ | [19,23,115,119] |
| 12 | $[0,\ -x,\ 0]$ | [20,22,116,118] |
| 13 | $\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | [25,32,121,128] |
| 14 | $\left[x+\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$ | [26,27,122,123] |
| 15 | $\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}-x\right]$ | [28,40,124,136] |
| 16 | $\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{3}{4}\right]$ | [29,39,125,135] |
| 17 | $\left[\frac{1}{4},\ x+\frac{3}{4},\ \frac{3}{4}\right]$ | [30,35,126,131] |
| 18 | $\left[\frac{1}{4},\ \frac{3}{4}-x,\ \frac{3}{4}\right]$ | [31,36,127,132] |
| 19 | $\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | [33,34,129,130] |
| 20 | $\left[\frac{1}{4}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | [37,38,133,134] |
| 21 | $\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [41,48,137,144] |
| 22 | $\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | [42,45,138,141] |
| 23 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | [43,47,139,143] |
| 24 | $\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [44,46,140,142] |
| 25 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | [49,56,145,152] |
| 26 | $\left[x+\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$ | [50,51,146,147] |
| 27 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}-x\right]$ | [52,64,148,160] |
| 28 | $\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{3}{4}\right]$ | [53,63,149,159] |
| 29 | $\left[\frac{3}{4},\ x+\frac{1}{4},\ \frac{3}{4}\right]$ | [54,59,150,155] |
| 30 | $\left[\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}\right]$ | [55,60,151,156] |
| 31 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | [57,58,153,154] |
| 32 | $\left[\frac{3}{4}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$ | [61,62,157,158] |
| 33 | $\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$ | [65,72,161,168] |
| 34 | $\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$ | [66,69,162,165] |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 35 | $\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$ | [67,71,163,167] |
| 36 | $\left[\frac{1}{2}, -x, \frac{1}{2}\right]$ | [68,70,164,166] |
| 37 | $\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$ | [73,80,169,176] |
| 38 | $\left[x+\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [74,75,170,171] |
| 39 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}-x\right]$ | [76,88,172,184] |
| 40 | $\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{4}\right]$ | [77,87,173,183] |
| 41 | $\left[\frac{3}{4}, x+\frac{3}{4}, \frac{1}{4}\right]$ | [78,83,174,179] |
| 42 | $\left[\frac{3}{4}, \frac{3}{4}-x, \frac{1}{4}\right]$ | [79,84,175,180] |
| 43 | $\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$ | [81,82,177,178] |
| 44 | $\left[\frac{3}{4}-x, \frac{3}{4}, \frac{1}{4}\right]$ | [85,86,181,182] |
| 45 | $\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$ | [89,96,185,192] |
| 46 | $\left[\frac{1}{2}, \frac{1}{2}, x\right]$ | [90,93,186,189] |
| 47 | $\left[\frac{1}{2}, \frac{1}{2}, -x\right]$ | [91,95,187,191] |
| 48 | $\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$ | [92,94,188,190] |

Table 7: Wyckoff site: 48g, site symmetry: ..21'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$ | [1,14,97,110] |
| 2 | $\left[\frac{3}{8}, y, y+\frac{1}{4}\right]$ | [2,57,98,153] |
| 3 | $\left[\frac{3}{8}, \frac{1}{2}-y, \frac{1}{4}-y\right]$ | [3,82,99,178] |
| 4 | $\left[\frac{1}{2}-y, y+\frac{1}{4}, \frac{1}{8}\right]$ | [4,93,100,189] |
| 5 | $\left[y, y+\frac{1}{4}, \frac{3}{8}\right]$ | [5,47,101,143] |
| 6 | $\left[\frac{1}{4}-y, \frac{3}{8}, \frac{1}{2}-y\right]$ | [6,46,102,142] |
| 7 | $\left[y+\frac{1}{4}, \frac{1}{8}, \frac{1}{2}-y\right]$ | [7,72,103,168] |
| 8 | $\left[\frac{1}{8}, -y, y+\frac{3}{4}\right]$ | [8,37,104,133] |
| 9 | $\left[\frac{7}{8}, y, y+\frac{3}{4}\right]$ | [9,50,105,146] |
| 10 | $\left[\frac{7}{8}, -y, \frac{1}{4}-y\right]$ | [10,75,106,171] |
| 11 | $\left[y+\frac{1}{4}, \frac{3}{8}, y\right]$ | [11,92,107,188] |
| 12 | $\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$ | [12,17,108,113] |
| 13 | $\left[\frac{1}{8}, \frac{1}{2}-y, y+\frac{1}{4}\right]$ | [13,32,109,128] |
| 14 | $\left[\frac{1}{2}-y, \frac{1}{4}-y, \frac{3}{8}\right]$ | [15,67,111,163] |
| 15 | $\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$ | [16,18,112,114] |
| 16 | $\left[-y, \frac{1}{4}-y, \frac{7}{8}\right]$ | [19,63,115,159] |
| 17 | $\left[y+\frac{3}{4}, \frac{7}{8}, y\right]$ | [20,83,116,179] |
| 18 | $\left[-y, y+\frac{3}{4}, \frac{1}{8}\right]$ | [21,76,117,172] |
| 19 | $\left[\frac{1}{4}-y, \frac{7}{8}, -y\right]$ | [22,30,118,126] |
| 20 | $\left[y, y+\frac{3}{4}, \frac{7}{8}\right]$ | [23,29,119,125] |
| 21 | $\left[y+\frac{3}{4}, \frac{1}{8}, -y\right]$ | [24,55,120,151] |
| 22 | $\left[\frac{1}{8}, y+\frac{1}{2}, \frac{3}{4}-y\right]$ | [25,38,121,134] |
| 23 | $\left[\frac{3}{8}, y+\frac{1}{2}, y+\frac{3}{4}\right]$ | [26,81,122,177] |
| 24 | $\left[\frac{3}{8}, -y, \frac{3}{4}-y\right]$ | [27,58,123,154] |
| 25 | $\left[\frac{1}{2}-y, y+\frac{3}{4}, \frac{5}{8}\right]$ | [28,69,124,165] |
| 26 | $\left[y+\frac{1}{4}, \frac{5}{8}, -y\right]$ | [31,96,127,192] |
| 27 | $\left[\frac{7}{8}, y+\frac{1}{2}, y+\frac{1}{4}\right]$ | [33,74,129,170] |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 28 | $\left[\frac{7}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$ | [34,51,130,147] |
| 29 | $\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$ | [35,68,131,164] |
| 30 | $\left[\frac{1}{4}-y,\ \frac{5}{8},\ y+\frac{1}{2}\right]$ | [36,41,132,137] |
| 31 | $\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{7}{8}\right]$ | [39,91,135,187] |
| 32 | $\left[y,\ \frac{3}{4}-y,\ \frac{5}{8}\right]$ | [40,42,136,138] |
| 33 | $\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$ | [43,87,139,183] |
| 34 | $\left[y+\frac{3}{4},\ \frac{3}{8},\ y+\frac{1}{2}\right]$ | [44,59,140,155] |
| 35 | $\left[-y,\ y+\frac{1}{4},\ \frac{5}{8}\right]$ | [45,52,141,148] |
| 36 | $\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$ | [48,79,144,175] |
| 37 | $\left[\frac{5}{8},\ y,\ \frac{3}{4}-y\right]$ | [49,62,145,158] |
| 38 | $\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$ | [53,95,149,191] |
| 39 | $\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$ | [54,94,150,190] |
| 40 | $\left[\frac{5}{8},\ -y,\ y+\frac{1}{4}\right]$ | [56,85,152,181] |
| 41 | $\left[\frac{3}{4}-y,\ \frac{1}{8},\ y+\frac{1}{2}\right]$ | [60,65,156,161] |
| 42 | $\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$ | [61,80,157,176] |
| 43 | $\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{5}{8}\right]$ | [64,66,160,162] |
| 44 | $\left[\frac{3}{4}-y,\ \frac{7}{8},\ \frac{1}{2}-y\right]$ | [70,78,166,174] |
| 45 | $\left[y+\frac{1}{2},\ y+\frac{3}{4},\ \frac{3}{8}\right]$ | [71,77,167,173] |
| 46 | $\left[\frac{5}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$ | [73,86,169,182] |
| 47 | $\left[\frac{3}{4}-y,\ \frac{5}{8},\ y\right]$ | [84,89,180,185] |
| 48 | $\left[y+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{8}\right]$ | [88,90,184,186] |

Table 8: Wyckoff site: 96h, site symmetry: 11'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1,97] |
| 2 | $\left[x+\frac{1}{4},\ \frac{1}{4}-z,\ y+\frac{1}{4}\right]$ | [2,98] |
| 3 | $\left[x+\frac{1}{4},\ z+\frac{1}{4},\ \frac{1}{4}-y\right]$ | [3,99] |
| 4 | $\left[z+\frac{1}{4},\ y+\frac{1}{4},\ \frac{1}{4}-x\right]$ | [4,100] |
| 5 | $\left[\frac{1}{4}-z,\ y+\frac{1}{4},\ x+\frac{1}{4}\right]$ | [5,101] |
| 6 | $\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$ | [6,102] |
| 7 | $\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$ | [7,103] |
| 8 | $[x,\ -y,\ -z]$ | [8,104] |
| 9 | $[-x,\ y,\ -z]$ | [9,105] |
| 10 | $[-x,\ -y,\ z]$ | [10,106] |
| 11 | $\left[y+\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$ | [11,107] |
| 12 | $\left[\frac{1}{4}-y,\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$ | [12,108] |
| 13 | $\left[\frac{1}{4}-x,\ z+\frac{1}{4},\ y+\frac{1}{4}\right]$ | [13,109] |
| 14 | $\left[\frac{1}{4}-x,\ \frac{1}{4}-z,\ \frac{1}{4}-y\right]$ | [14,110] |
| 15 | $\left[z+\frac{1}{4},\ \frac{1}{4}-y,\ x+\frac{1}{4}\right]$ | [15,111] |
| 16 | $\left[\frac{1}{4}-z,\ \frac{1}{4}-y,\ \frac{1}{4}-x\right]$ | [16,112] |
| 17 | $[z,\ x,\ y]$ | [17,113] |
| 18 | $[y,\ z,\ x]$ | [18,114] |
| 19 | $[-y,\ z,\ -x]$ | [19,115] |
| 20 | $[-z,\ -x,\ y]$ | [20,116] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 21 | $[-y, -z, x]$ | [21,117] |
| 22 | $[z, -x, -y]$ | [22,118] |
| 23 | $[y, -z, -x]$ | [23,119] |
| 24 | $[-z, x, -y]$ | [24,120] |
| 25 | $\left[x, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [25,121] |
| 26 | $\left[x+\frac{1}{4}, \frac{3}{4}-z, y+\frac{3}{4}\right]$ | [26,122] |
| 27 | $\left[x+\frac{1}{4}, z+\frac{3}{4}, \frac{3}{4}-y\right]$ | [27,123] |
| 28 | $\left[z+\frac{1}{4}, y+\frac{3}{4}, \frac{3}{4}-x\right]$ | [28,124] |
| 29 | $\left[\frac{1}{4}-z, y+\frac{3}{4}, x+\frac{3}{4}\right]$ | [29,125] |
| 30 | $\left[\frac{1}{4}-y, x+\frac{3}{4}, z+\frac{3}{4}\right]$ | [30,126] |
| 31 | $\left[y+\frac{1}{4}, \frac{3}{4}-x, z+\frac{3}{4}\right]$ | [31,127] |
| 32 | $\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$ | [32,128] |
| 33 | $\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$ | [33,129] |
| 34 | $\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [34,130] |
| 35 | $\left[y+\frac{1}{4}, x+\frac{3}{4}, \frac{3}{4}-z\right]$ | [35,131] |
| 36 | $\left[\frac{1}{4}-y, \frac{3}{4}-x, \frac{3}{4}-z\right]$ | [36,132] |
| 37 | $\left[\frac{1}{4}-x, z+\frac{3}{4}, y+\frac{3}{4}\right]$ | [37,133] |
| 38 | $\left[\frac{1}{4}-x, \frac{3}{4}-z, \frac{3}{4}-y\right]$ | [38,134] |
| 39 | $\left[z+\frac{1}{4}, \frac{3}{4}-y, x+\frac{3}{4}\right]$ | [39,135] |
| 40 | $\left[\frac{1}{4}-z, \frac{3}{4}-y, \frac{3}{4}-x\right]$ | [40,136] |
| 41 | $\left[z, x+\frac{1}{2}, y+\frac{1}{2}\right]$ | [41,137] |
| 42 | $\left[y, z+\frac{1}{2}, x+\frac{1}{2}\right]$ | [42,138] |
| 43 | $\left[-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$ | [43,139] |
| 44 | $\left[-z, \frac{1}{2}-x, y+\frac{1}{2}\right]$ | [44,140] |
| 45 | $\left[-y, \frac{1}{2}-z, x+\frac{1}{2}\right]$ | [45,141] |
| 46 | $\left[z, \frac{1}{2}-x, \frac{1}{2}-y\right]$ | [46,142] |
| 47 | $\left[y, \frac{1}{2}-z, \frac{1}{2}-x\right]$ | [47,143] |
| 48 | $\left[-z, x+\frac{1}{2}, \frac{1}{2}-y\right]$ | [48,144] |
| 49 | $\left[x+\frac{1}{2}, y, z+\frac{1}{2}\right]$ | [49,145] |
| 50 | $\left[x+\frac{3}{4}, \frac{1}{4}-z, y+\frac{3}{4}\right]$ | [50,146] |
| 51 | $\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{3}{4}-y\right]$ | [51,147] |
| 52 | $\left[z+\frac{3}{4}, y+\frac{1}{4}, \frac{3}{4}-x\right]$ | [52,148] |
| 53 | $\left[\frac{3}{4}-z, y+\frac{1}{4}, x+\frac{3}{4}\right]$ | [53,149] |
| 54 | $\left[\frac{3}{4}-y, x+\frac{1}{4}, z+\frac{3}{4}\right]$ | [54,150] |
| 55 | $\left[y+\frac{3}{4}, \frac{1}{4}-x, z+\frac{3}{4}\right]$ | [55,151] |
| 56 | $\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$ | [56,152] |
| 57 | $\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$ | [57,153] |
| 58 | $\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$ | [58,154] |
| 59 | $\left[y+\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}-z\right]$ | [59,155] |
| 60 | $\left[\frac{3}{4}-y, \frac{1}{4}-x, \frac{3}{4}-z\right]$ | [60,156] |
| 61 | $\left[\frac{3}{4}-x, z+\frac{1}{4}, y+\frac{3}{4}\right]$ | [61,157] |
| 62 | $\left[\frac{3}{4}-x, \frac{1}{4}-z, \frac{3}{4}-y\right]$ | [62,158] |
| 63 | $\left[z+\frac{3}{4}, \frac{1}{4}-y, x+\frac{3}{4}\right]$ | [63,159] |
| 64 | $\left[\frac{3}{4}-z, \frac{1}{4}-y, \frac{3}{4}-x\right]$ | [64,160] |
| 65 | $\left[z+\frac{1}{2}, x, y+\frac{1}{2}\right]$ | [65,161] |
| 66 | $\left[y+\frac{1}{2}, z, x+\frac{1}{2}\right]$ | [66,162] |
| 67 | $\left[\frac{1}{2}-y, z, \frac{1}{2}-x\right]$ | [67,163] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 68 | $\left[\frac{1}{2}-z,\,-x,\,y+\frac{1}{2}\right]$ | [68,164] |
| 69 | $\left[\frac{1}{2}-y,\,-z,\,x+\frac{1}{2}\right]$ | [69,165] |
| 70 | $\left[z+\frac{1}{2},\,-x,\,\frac{1}{2}-y\right]$ | [70,166] |
| 71 | $\left[y+\frac{1}{2},\,-z,\,\frac{1}{2}-x\right]$ | [71,167] |
| 72 | $\left[\frac{1}{2}-z,\,x,\,\frac{1}{2}-y\right]$ | [72,168] |
| 73 | $\left[x+\frac{1}{2},\,y+\frac{1}{2},\,z\right]$ | [73,169] |
| 74 | $\left[x+\frac{3}{4},\,\frac{3}{4}-z,\,y+\frac{1}{4}\right]$ | [74,170] |
| 75 | $\left[x+\frac{3}{4},\,z+\frac{3}{4},\,\frac{1}{4}-y\right]$ | [75,171] |
| 76 | $\left[z+\frac{3}{4},\,y+\frac{3}{4},\,\frac{1}{4}-x\right]$ | [76,172] |
| 77 | $\left[\frac{3}{4}-z,\,y+\frac{3}{4},\,x+\frac{1}{4}\right]$ | [77,173] |
| 78 | $\left[\frac{3}{4}-y,\,x+\frac{3}{4},\,z+\frac{1}{4}\right]$ | [78,174] |
| 79 | $\left[y+\frac{3}{4},\,\frac{3}{4}-x,\,z+\frac{1}{4}\right]$ | [79,175] |
| 80 | $\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,-z\right]$ | [80,176] |
| 81 | $\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,-z\right]$ | [81,177] |
| 82 | $\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,z\right]$ | [82,178] |
| 83 | $\left[y+\frac{3}{4},\,x+\frac{3}{4},\,\frac{1}{4}-z\right]$ | [83,179] |
| 84 | $\left[\frac{3}{4}-y,\,\frac{3}{4}-x,\,\frac{1}{4}-z\right]$ | [84,180] |
| 85 | $\left[\frac{3}{4}-x,\,z+\frac{3}{4},\,y+\frac{1}{4}\right]$ | [85,181] |
| 86 | $\left[\frac{3}{4}-x,\,\frac{3}{4}-z,\,\frac{1}{4}-y\right]$ | [86,182] |
| 87 | $\left[z+\frac{3}{4},\,\frac{3}{4}-y,\,x+\frac{1}{4}\right]$ | [87,183] |
| 88 | $\left[\frac{3}{4}-z,\,\frac{3}{4}-y,\,\frac{1}{4}-x\right]$ | [88,184] |
| 89 | $\left[z+\frac{1}{2},\,x+\frac{1}{2},\,y\right]$ | [89,185] |
| 90 | $\left[y+\frac{1}{2},\,z+\frac{1}{2},\,x\right]$ | [90,186] |
| 91 | $\left[\frac{1}{2}-y,\,z+\frac{1}{2},\,-x\right]$ | [91,187] |
| 92 | $\left[\frac{1}{2}-z,\,\frac{1}{2}-x,\,y\right]$ | [92,188] |
| 93 | $\left[\frac{1}{2}-y,\,\frac{1}{2}-z,\,x\right]$ | [93,189] |
| 94 | $\left[z+\frac{1}{2},\,\frac{1}{2}-x,\,-y\right]$ | [94,190] |
| 95 | $\left[y+\frac{1}{2},\,\frac{1}{2}-z,\,-x\right]$ | [95,191] |
| 96 | $\left[\frac{1}{2}-z,\,x+\frac{1}{2},\,-y\right]$ | [96,192] |